Department of Economics

MSc Extended Essay Cover Sheet

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Financial Repression in a New Keynesian Model with Long-Term Bonds and Consumption Habits

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Financial Repression in a New Keynesian Model with Long-Term Bonds and Consumption Habits

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Abstract

This essay assesses the cost of financial repression within a simple version of the New Keynesian model (NK) with long-term government bonds, external consumption habits, and sticky prices, according to Calvo. Using the Ramsey optimal policy, I find that the central bank can neutralize a fiscal shock either via an immediate change in ex-post real return or a change in current and future ex-ante real interest rate. In the absence of habits, loss from financial repression is minimized when all debt is rolled over periodically or when all government debt comprises perpetuities. Financial repression is more costly in general as habits become more persistent, which disables the ex-ante real interest rate channel, and minimal loss cannot be achieved with period rollover. However, a minimal loss is still attainable under a perpetuity debt structure.

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1 Introduction

In the context of recent economic turmoil, the relevance of financial repression, governments' or central banks' attempt to reduce the real value of government borrowing, is surging. The Covid-19 pandemic, among other factors, has caused an unprecedented increase in public debt, with the United States experiencing a 30% year-over-year increase in debt-to-GDP ratio in the second quarter of 2020 (Federal Reserve Bank of St. Louis, U.S. Office of Management and Budget, 2023). With such substantial government debt increases, relying solely on future tax hikes may not be feasible, prompting authorities to explore alternative options like financial repression.

This essay examines the welfare implications of financial repression within a new Keynesian model featuring monopolistic competition and Calvo pricing, introducing monopoly pricing distortions and nominal rigidities. Household optimization behavior is governed by external consumption habits anchoring individual consumption to the past cross-sectional average. This feature captures empirical persistence in consumption patterns. Consequently, the model features a backward-looking PC and IS curve. Further, the essay deviates from standard NK models by introducing a market for long-term government bonds. This prevents an optimal solution using unrealistically aggressive short-term policy choices that theoretically absorb fiscal shocks instantaneously without long-term impact.

It is worth noting that in the absence of consumption habits, the essay finds that the central bank's loss-minimizing Ramsey policy employs financial repression by reducing the nominal interest rate and yield below the steady-state level. This decrease in interest rates lowers the current and future real interest rates, enabling the government to refinance expiring debt at a lower cost. However, this comes at a loss, which is magnified as persistence in the consumption process is introduced. Even for low habit persistence, the ex-post real return channel is the more credible and cost-minimizing form of financial repression. In a more realistic setting where debt maturity is a choice variable, minimizing loss increases maturity, which aligns with empirical findings of a Macaulay duration close to one.

The essay relates to a flourishing field of research on calibrating DSGE models and estimating the impulse response functions in response to shocks to the economy (Smets & Wouters, 2003). Additionally, it touches upon the relevance of consumption habits in modeling a new Keynesian economy (Ravn et al., 2006) with the aim of quantifying welfare implications (Amato & Laubach, 2004; Leith et al., 2012) of financial repression.

The remainder of this essay comprises four sections: Section 2 presents the theoretical model. Section 3 discusses the parameter calibration underpinning the Ramsey policy. Subsequently, section 4 elaborates on the results of the simulations, examining the welfare effects of monetary policy interventions in response to a fiscal shock. Finally, section 5 concludes.

2 Model with External Habits

2.1 The Household Sector

The specification for the household problem follows Leith et al. (2012), Dennis (2008), and Ravn et al. (2006).

The economy consists of a continuum of households $k \in [0, 1]$, receiving utility from consuming a habit-adjusted composite good X_t^k and incurring a disutility from labor provision N_t^k. Habits are superficial, additive, and external. Additive superficial habits occur on the aggregate consumption good C_t^k level and are added to the current household consumption. Hence $X_t^k = C_t^k - \zeta C_{t-1}$, where $C_t^k = \left(\int_0^1 C_t^k(\iota)^{\frac{\epsilon-1}{\epsilon}} d\iota\right)^{\frac{\epsilon}{\epsilon-1}}$ is the aggregate over the continuum of goods that household k consumes, i.e., the household's consumption basket of differentiated goods. $\epsilon > 1$ is the elasticity of substitution between differentiated goods, and ζ is the persistence of habit formation. $C_{t-1} = \int_0^1 C_{t-1}^k dk$ [1] represents the cross-sectional average consumption basket, and $C_t(\iota) = \int_0^1 C_t^k(\iota) dk$ is the cross-sectional average consumption of a good ι. The lag ζC_{t-1} in the habit-adjusted composite good is the external habit stock, which encompasses external habit formation according to the "keeping up with the Joneses"

[1] In the following, the same aggregation technique is assumed to hold for variables indicated k

specification (Abel, 1990). Setting $\zeta = 0$ returns the standard model without habits. The central feature here is that households desire a consumption level relative to what others are consuming, which introduces a consumption externality whereby households overconsume, not accounting for the increase in aggregate consumption due to their individual consumption, which is reflected in a higher habit stock in the following period.

Additive habits can capture the notion of a subsistence level: consumption below the general level decreases utility by a lot (since the Inada condition holds, the marginal utility of habit-adjusted consumption approaches infinity as individual consumption approaches the aggregate level). The consequence is persistence in the process for optimal consumption.

2.1.1 Expenditure Minimization

Given the price for differentiated goods $P_t(\iota)$, the cost of the average household's consumption basket is $P_t = \min_{C_t^k(\iota)} \int_0^1 C_t^k(\iota) P_t(\iota) d\iota$. This allows solving the household's expenditure minimization problem, which yields the individual expenditure-minimizing demand function for differentiated good ι:

$$C_t^k(\iota) = \left(\frac{P_t(\iota)}{P_t}\right)^{-\epsilon} C_t^k \tag{1}$$

at a price level

$$P_t = \left(\int_0^1 P_t(\iota)^{1-\epsilon} d\iota\right)^{\frac{1}{1-\epsilon}} \tag{2}$$

Aggregating over households yields the aggregate demand for good ι:

$$C_t(\iota) = \int_0^1 C_t^k(\iota) dk = \left(\frac{P_t(\iota)}{P_t}\right)^{-\epsilon} C_t \tag{3}$$

Because habits are superficial, this demand function is identical to the case of no habit formation, implying that the firm continues to face a static price-quantity decision.

2.1.2 Utility Maximization

The flip side of the household problem is utility maximization: households aim to maximize the present discounted value of lifetime utility subject to an intertemporal budget constraint. I assume additively separable utility with habit-adjusted consumption and disutility of labor and the government bond market serving as the sole savings vehicle. The problem can be formulated as follows:

$$\max_{\{N_{t+l}^k, X_{t+l}^k, B_{t+l}^k\}_{l=0}^{\infty}} \sum_{l=0}^{\infty} \beta^l E_t \left(\frac{\left(X_{t+l}^k\right)^{\left(1-\frac{1}{\sigma}\right)} - 1}{1-\frac{1}{\sigma}} - \frac{\left(N_{t+l}^k\right)^{\left(1+\frac{1}{\psi}\right)}}{1+\frac{1}{\psi}} \right)$$

subject to the intertemporal budget constraint:

$$C_t^k + \frac{Q_t B_t^k}{P_t} = \frac{W_t N_t^k}{P_t} + \frac{(1+\delta Q_{t+1}) B_{t-1}^k}{P_t} + \frac{D_t^k}{P_t}$$

where σ is the intertemporal elasticity of substitution, ψ is the Frisch Elasticity of labor supply, W_t is the nominal wage, B_t^k is the quantity of government bonds held by household k, D_t^k represents dividends from firms net of lump-sum transfers from the government and taxes, and Q_t is the nominal bond price.

Solving the utility maximization problem yields two optimality conditions: the Euler Equation characterizes the intertemporal trade-off between habit-adjusted consumption today and tomorrow:

$$\left(X_t^k\right)^{-\frac{1}{\sigma}} = \beta E_t \left(\frac{(1+\delta Q_{t+1}) P_t}{P_{t+1} Q_t} \left(X_{t+1}^k\right)^{-\frac{1}{\sigma}} \right) \tag{4}$$

And the consumption-leisure trade-off:

$$\frac{\left(N_t^k\right)^{\frac{1}{\psi}}}{\left(X_t^k\right)^{-\frac{1}{\sigma}}} = w_t, \quad \text{defining the real wage } w_t = \frac{W_t}{P_t} \tag{5}$$

Since all households k have the same objective and the same optimality conditions, house-

holds are symmetric, and we can use the cross-sectional average $(\cdot)_t$ of any variable $(\cdot)_t^k$ as the representative household.

2.1.3 Bond Market

In the basic New Keynesian model, all bonds are rolled over every period. However, in a more realistic scenario, bonds can have maturity longer than a single period. Suppose a newly issued bond paying a sequence of coupon payments $1, \delta, \delta^2, \delta^3, \ldots$ in periods $t, t+1, t+2, \ldots$ after issuance, where $\delta \in [0, 1]$ (a perpetual bond with geometrically decaying coupon payments).

The nominal bond price Q_t of a bond issued in period t is given by:

$$Q_t = \frac{1}{1+j_t} + \frac{\delta}{(1+j_t)^2} + \frac{\delta^2}{(1+j_t)^3} + \ldots = (1+j_t)^{-1} \sum_{s=0}^{\infty} \left(\frac{\delta}{1+j_t}\right)^s = \frac{1}{1+j_t-\delta} \tag{6}$$

Define the ex-post real return R_t on a bond between period t and $t+1$. A bond worth Q_t in period t will be worth δQ_{t+1} in $t+1$, as the bondholder receives a coupon of 1 and has a stream of future coupons worth $\frac{\delta}{1+j_t} + \frac{\delta^2}{(1+j_t)^2} + \frac{\delta^3}{(1+j_t)^3}$, which is equivalent to δQ_{t+1}. Hence, the following identity holds between time periods t and $t+1$:

$$(1+R_{t+1})Q_t = \frac{(1+\delta Q_{t+1})P_t}{P_{t+1}} \tag{7}$$

By rearranging and noticing that $(1+\pi_t) = \frac{P_t}{P_{t-1}}$, I obtain:

$$(1+R_{t+1}) = \frac{(1+j_t)}{(1+\pi_{t+1})} \frac{(1-\frac{\delta}{1+j_t})}{(1-\frac{\delta}{1+j_{t+1}})}$$

If $\delta = 0$, we return to the standard New Keynesian model, where the equivalent of the ex-post real return on bonds is the short-term real interest rate r_t. This result arises from setting $\delta = 0$ and imposing $j_t = i_t$ (when all bonds are single-period bonds, the nominal interest rate is effectively the bond yield since $Q_t = \frac{1}{1+i_t}$). Recalling the Fisher equation $(1+r_t) = \frac{(1+i_t)}{(1+\pi_{t+1})}$

and noticing that no-arbitrage between bonds of different maturity requires, according to the consumption Euler equation: $(1 + E_t R_{t+1}) = (1 + r_t)$ Otherwise, consumers would strictly invest in the vehicle with a higher return (disregarding the limited supply of bonds). Hence, there would only exist a bond market for single-period bonds or a market for long-term bonds.

2.2 The Firm

Via the firm side, the economy faces two types of distortions: monopolistic competition and sticky prices following the Calvo (1983) specification. The setup is standard following, for instance, Galí (2008).

A continuum of firms on a unit interval $[0, 1]$ is assumed to produce a continuum of goods. Each firm $\iota \in [0, 1]$ is the unique producer of a differentiated good ι, giving rise to monopolistic competition. Consequently, firms set prices with a markup, introducing a monopoly pricing externality through underproduction. Further, each firm faces a standard neoclassical production function (constant return to scale, decreasing returns to factors, Inada condition holds) with firm-specific fixed capital [2]:

$$Y_t(\iota) = A_t N_t(\iota)^{1-\alpha} \tag{8}$$

where A_t is the exogenous productivity level, and $\alpha \in [0, 1)$ is the labor share of output. To produce output $Y_t(\iota)$, the firm needs $\left(\frac{Y_t(\iota)}{A_t}\right)^{\frac{1}{1-\alpha}}$ units of labor.

[2] Capital is not shown for simplicity

2.2.1 Production Decision

In a perfectly competitive labor market[3] where firms take the wage as given, the real cost of production is thus:

$$Z_t(Y_t(\iota)) = w_t \left(\frac{Y_t(\iota)}{A_t}\right)^{\frac{1}{1-\alpha}} \tag{9}$$

and thus, denoting $\eta \equiv \frac{\alpha}{1-\alpha}$, real marginal costs are:

$$z_t = \frac{(1+\eta)w_t}{A_t}\left(\frac{Y_t(\iota)}{A_t}\right)^{\eta} \tag{10}$$

Good Market Clearing

Under firm-level market clearing where consumption is the sole source of demand, $Y_t(\iota) = C_t(\iota)$. This must hold for all ι, thus $C_t = Y_t$. By taking the firm's production function, substituting household demand, and aggregating over $N_t(\iota)$, we get the goods market solution:

$$Y_t = \frac{N^{1/(1+\eta)}A_t}{\Delta_t} \tag{11}$$

with relative-price distortions $\Delta_t = \left(\int_0^1 \left(\frac{P_t(\iota)}{P_t}\right)^{-\epsilon(1+\eta)} d\iota\right)^{\frac{1}{1+\eta}}$

While the firm's production function links firm-specific output to firm-level employment, this is not necessarily true on an aggregate level. Instead, the demand function for individual goods $Y_t(\iota) = \left(\frac{P_t(\iota)}{P_t}\right)^{-\epsilon} Y_t$ is used as a link.

Labor Market Clearing

Labor markets clear. Equating household labor supply with firm labor demand and

[3]Despite households providing differentiated labor $N_t^k(\iota)$, here I assume, all provide the same quality of labor for simplicity.

imposing the goods market solution, the optimal wage is:

$$w_t = (Y_t - \zeta Y_{t-1})^{\frac{1}{\sigma}} \left(\frac{Y_t \Delta_t}{A_t} \right)^{\frac{1+\eta}{\psi}} \tag{12}$$

2.2.2 Pricing Decision

In addition, the model assumes the presence of sticky prices according to Calvo (1983), which is a time-dependent pricing model without synchronization. The Calvo parameter θ is the externally given likelihood of an individual firm not being able to change prices in a given period. When setting prices, firms take real wages as given and maximize real profits, acknowledging price stickiness.

Flexible Price Firm

Given the monopolistic competition setup, firms have price-setting power as the sole producer of good ι. Thus, the hypothetically fully flexible profit-maximizing monopolist optimally equates marginal cost and marginal revenue. Given the demand curve from the household section, it is optimal for the monopolist to engage in markup pricing charging a fixed markup $\mu = \frac{\epsilon}{\epsilon - 1}$ above marginal cost. Denoting the fully flexible price as $\widetilde{P}_t$ and recalling the marginal cost equation, such firms choose:

$$\frac{\widetilde{P}_t}{P_t} = \mu \frac{(1+\eta) w_t}{A_t} \left(\frac{Y_t}{A_t} \right)^{\eta} \left(\frac{\widetilde{P}_t}{P_t} \right)^{-\epsilon\eta} \tag{13}$$

Sticky Price Firm

The sticky price firm solves the profit maximization problem. Whenever an exogenous price change opportunity occurs, all firms changing prices in period t choose an identical optimal reset price P_t^*, which will hold until the next price change. Thus, in an infinite horizon problem, P_t^* solves the first-order condition of the profit maximization, which amounts to:

$$E_t \sum_{l=0}^{\infty} \theta^l q_{t+l|t} Y_{t+l} \left(\frac{P_t^*}{P_{t+l}}\right)^{-\epsilon} \left(\frac{P_t^*}{P_{t+l}} - \mu \left(\frac{P_t^*}{P_{t+l}}\right)^{-\epsilon\eta} z_{t+l}\right) = 0 \tag{14}$$

where $q_{t+l|t} = q_t \dots q_{t+l-1}$ is the real discount factor such that $q_t = Q_t \left(\frac{P_{t+1}}{P_t}\right)$.

Solving the problem yields that the optimal reset price is a forward-looking weighted average of current and future optimal flexible prices. Then, the aggregate price level under sticky prices is:

$$P_t = \left((1-\theta) \sum_{l=0}^{\infty} \theta^l (P^*)_{t-l}^{1-\epsilon}\right)^{\frac{1}{1-\epsilon}} \tag{15}$$

Finally, relative-price distortions are:

$$\Delta_t = \left((1-\theta) \sum_{l=0}^{\infty} \theta^l \left(\frac{P_{t-l}^*}{P_t}\right)^{-\epsilon(1+\eta)}\right)^{\frac{1}{1+\eta}} \tag{16}$$

2.3 The Government

The government issues bonds and collects taxes subject to the flow budget constraint:

$$T_t + \frac{Q_t B_t}{P_t} = \frac{(1+\delta Q_t) B_{t-1}}{P_t} \tag{17}$$

The setup captures the idea that tax revenue or transfers and the value of current bonds must equate to the current value of future bonds. The government cannot let the debt level explode and run a long-term deficit for the infinite future.

Rewriting $D_t = \frac{Q_t B_t}{P_t}$ for the real value of government liabilities at the end of period t, we have:

$$T_t + D_t = (1+R_t) D_{t-1} \tag{18}$$

Further, I assume fiscal policy in the form of a wage bill subsidy to neutralize the monopoly pricing externality.

2.4 Linearized Model

Log-linearizing the optimality conditions[4] from households, firms, and the government and assuming constant productivity $A_t = A$ yields the log-linearized equations. [5]

$$\pi_t = \beta E_t \pi_{t+1} + \xi\nu \left(\left[\frac{(1+\eta)}{\psi} + \eta \right] y_t + \frac{(y_t - \zeta y_{t-1})}{\sigma(1-\zeta)} \right) \tag{19}$$

$$y_t = \frac{1}{1+\zeta} E_t y_{t+1} + \frac{\zeta}{1+\zeta} y_{t-1} - \frac{\sigma(1-\zeta)}{1+\zeta} r_t \tag{20}$$

$$r_t = E_t R_{t+1} \tag{21}$$

$$j_t = \beta\delta E_t j_{t+1} + (1-\beta\delta) i_t \tag{22}$$

$$R_t = j_{t-1} - \pi_t - \frac{\beta\delta}{(1-\beta\delta)}(j_t - j_{t-1}) \tag{23}$$

$$\beta d_t = d_{t-1} + R_t + \beta\phi_t \tag{24}$$

2.4.1 NKPC and IS Curve

Phillips Curve

The habit-adjusted New Keynesian Phillips Curve (NKPC) relates inflation π_t to the output gap y_t. In the special case without habit formation, $\zeta = 0$, the NKPC reduces to the standard forward-looking version $\pi_t = \beta E_t \pi_{t+1} + \kappa y_t$ where κ is the Phillips Curve gradient linking together firm-level real rigidities, aggregate real rigidities and nominal rigidities (inversely related to ν, γ, ξ respectively) such that

$$\kappa = \nu\gamma\xi \quad \nu = \frac{1}{1+\epsilon\eta} \quad \gamma = \frac{1}{\sigma} + \frac{1+\eta}{\psi} + \eta \quad \xi = \frac{(1-\theta)(1-\beta\theta)}{\theta}$$

Conversely, habits introduce a backward-looking inflation process, which arises exclu-

[4] Log-linearization in Appendix A

[5] For ease of notation, I drop the ' $\widehat{}$ ' notation in the linearized model

sively from the elasticity of marginal cost with respect to the output gap. The elasticity of inflation with respect to changes in marginal cost remains unaffected by habit formation. This results from the fact that nominal rigidities are governed by the Calvo parameter, and firm-level real rigidities are unaffected by habit formation because the firm continues to face a static aggregate demand function. Real marginal costs are affected by the dynamic nature of households' labour supply. That is, external habits render optimal labour-supply dynamic because optimal consumption is also a function of yesterday's aggregate consumption, which is a function of yesterday's labour supply. Hence, the inflation process exhibits greater inertia with habit formation.

IS Curve

The habit-adjusted IS equation links the ex-ante real interest rate r_t to the output gap y_t. Again when $\zeta = 0$, the economy returns to the standard forward-looking IS equation $y_t = E_t y_{t+1} - \sigma r_t$. Under external habits, however, the output gap is the weighted average of past and expected future output gaps. The new backward-looking component captures persistence in the optimal consumption process, which implies persistence in the output gap under market clearing. Additionally, in the absence of fluctuations in the natural rate of interest, the intertemporal elasticity of substitution σ in the no-habit case implies that a one percentage point increase in ex-ante real interest rate is associated with a σ percentage point decrease in the output gap. Under $\zeta \neq 0$, however, the interest elasticity of the output gap and consumption depends also on the habit parameter. Precisely, with external habits, the link is weakened, and an equivalent increase in real interest is associated with a lesser decrease in the output gap, ceteris paribus. This seems intuitive since habits introduce persistence in the optimal consumption choice. Understanding the real interest rate as the cost of transferring consumption units from tomorrow to today, the representative household is less responsive to a change in cost because of its desire to meet the level of external habit stock.

2.4.2 Bond Market

The linearized bond-market equations are independent of habit formation. Mathematically, in the optimization problem, habit-adjusted consumption but also standard non-adjusted consumption drop out.

Bond Yield

Iterating forward the bond yield (EQ 22) starting from t_0 and taking expectations

$$j_t = (1-\beta\delta)\left(i_{t_0} + \beta\delta E_{t_0} i_{t_0+1} + (\beta\delta)^2 E_{t_0} i_{t_0+2} + \ldots\right) + \lim_{t\to\infty} E_{t_0}\left((\beta\delta)^t j_{t_0+t}\right)$$

$$j_t = (1-\beta\delta)\sum_{t=0}^{\infty}(\beta\delta)^t E_{t_0} i_{t_0+t}$$

This illustrates that the bond yield behaves similarly to an expectations theory of interest rates style interpretation. That is, the current bond yield is a weighted average of current and expected future nominal interest rates. Looking at the weights, we see that the weights depend on β and δ. As β increases, weights go down less rapidly, and future rates are more relevant to the present if people are not discounting the future very much. As δ increases, weights also go down less rapidly: with a longer bond duration, current yields are related to expected interest rates further in the future. Then, in fact, both β and δ can be thought of as a discount factor, with the only subtlety that δ captures the chance of not having rolled over the bond by the time the future arrives.

Ex-Post Real Return

The ex-post real return on bonds captures the standard inverse yield-price relationship: when yield increases, the differential between yield in t and yield in $t-1$ is positive, which decreases real return given the negative coefficient $-\beta\delta/(1-\beta\delta)$.

$$R_t = j_{t-1} - \pi_t - \frac{\beta\delta}{1-\beta\delta}(j_t - j_{t-1})$$

2.4.3 Government

In EQ(24), the fiscal shock is defined as a deviation of tax returns from their steady-state relative to the steady-state debt level $\phi_t = \frac{(T-T_t)}{D}$. Hence, the fiscal constraint is motivated by the idea that increased debt or decreased current tax revenue as a result of a sizable fiscal shock can only be compensated by increasing future taxes to a limited extent since the budget constraint must hold. Furthermore, monetary policy is linked to the fiscal side via its impact on inflation, which in turn affects the ex-post real value of debt repayments, and additionally through the ex-ante real interest rate, which is within the control of the central bank under sticky prices. We also find that in steady-state, monetary policy has no effect on the real value of debt, so $\pi = 0$ is possible.

For intuition, we can iterate forward starting from t_0:

$$d_{t-1} = \beta d_t - R_t - \beta\phi_t$$

$$= \beta(\beta E_t d_{t+1} - E_t R_{t+1} - \beta E_t \phi_{t+1}) - R_t - \beta\phi_t$$

$$= [\ldots]$$

$$d_{t_0-1} = \lim_{t\to\infty} \beta^{t+1} E_{t_0} d_{t_0+t} - \sum_{t=0}^{\infty} \beta^t E_{t_0} R_{t_0+t} - \beta\phi_{t_0}$$

This makes use of the fact that the fiscal shock follows a mean-zero i.i.d. process, implying that in expectation $E_{t_0}\phi_t = 0$ for all $t > t_0$. Since the economy is in steady-state before the shock hits at t_0, we know $d_{t_0-1} = 0$. Then, imposing the transversality condition (TVC) $\lim_{t\to\infty} \beta^t d_{t_0+t} = 0$ and substituting for the ex-post real return on bonds, we find:

$$-\beta\phi_{t_0} = R_{t_0} + \sum_{t=0}^{\infty} \beta^{t+1} E_{t_0} r_{t_0+t}$$

$$-\phi_{t_0} = \beta^{-1} R_{t_0} + \sum_{t=0}^{\infty} \beta^t E_{t_0} r_{t_0+t}$$

Recalling that ϕ_t is an i.i.d. shock today, which is not followed by a counter-shock

tomorrow, we can interpret ϕ_t as the exogenously given size of the fiscal shock that cannot be compensated by future changes in taxes. Then, for the budget identity to hold, a positive fiscal shock (tax returns below steady state, like Covid) at t_0 can be counteracted only by either decreasing the ex-post real return on bonds today (R_{t_0}) or lowering ex-ante.

2.4.4 Loss Function

Together we have a New Keynesian economy with six log-linearized equations for seven unknowns, leaving one degree of freedom to solve for the policy rate i_t. These serve as the constraints to the optimal policy problem (Ramsey problem), with the objective of minimizing the utility-based loss function [6].

$$L_t = -\frac{1}{2}\Omega \sum_{l=0}^{\infty} \beta^l E_t \left\{ \frac{\epsilon}{\upsilon\xi}\pi_t^2 + \left(\omega + \frac{1+\eta}{\psi} + \eta\right) y_t^2 - 2\zeta\omega y_t y_{t-1} + \zeta^2 \omega y_{t-1}^2 \right\}$$

where $\omega = \frac{1}{\sigma(1-\beta\zeta)(1-\zeta)}$ and $\Omega = \bar{N}^{1+\frac{1}{\psi}}$.

The loss function is constructed from a second-order Taylor approximation of the utility U_t around the steady state, thus measuring welfare loss from monetary policy as a fraction of steady-state consumption. Unlike the standard NK model, external habits introduce an additional externality: overconsumption due to the habit-stock target. Hence, it is difficult to write the loss function in the usual output gap notation. Even under fully flexible prices, when output is at its natural level, a wedge between natural and optimal output remains (Leith et al., 2012). However, one can show that the minimization of the output gap is similar to minimizing output deviations from the social planner optimum[7]. Thus I use the above as a simplified alternative. In the special case without external habits ($\zeta = 0$), the loss function reduces to the standard summation over current:

[6] Derivation in Appendix C

[7] See appendix B for the social planner problem and appendix C for the derivation of the output deviation from the natural level

$$L_t = -\frac{1}{2}\Omega \sum_{l=0}^{\infty} \beta^l E_t \left\{ \frac{\epsilon}{v\xi}\pi_t^2 + \gamma y_t^2 \right\}$$

The loss function captures a trade-off between output and inflation, where the relative importance to the social planner is captured by the weighing parameters for inflation and the output gap.

No habits: In the no-habit model, a non-zero output gap is penalized depending on the level of aggregate-level real rigidities as measured by γ^{-1}, where γ is the elasticity of real marginal cost with respect to aggregate output. The penalty for inflation depends on the price elasticity of individual demand curves for differentiated goods ϵ, the extent of firm-level real rigidities ν^{-1}, and nominal rigidities ξ^{-1}. Hence, higher nominal rigidities and higher firm-level real rigidities increase welfare loss through inflation, and lower aggregate real rigidities increase welfare loss through the output gap.

Habits: From the loss function, it becomes clear that the divine coincidence steady-state of $\pi = 0$ and $y = 0$ is efficient. Furthermore, interest rates as the main policy tool do not appear in the loss function and are only a means to achieve a desired output gap and inflation.

3 Parameter Calibration

The model consists of seven parameters: $\bar{r}, \sigma, \epsilon, \psi, \eta, \theta$, and δ. The calibration is conducted for the US, with certain parameters being more widely available on a non-country-specific level, as indicated below.

Steady-State Real Interest Rate ($\bar{r}$)

The US steady-state real interest rate is commonly estimated to be approximately 2% p.a. (Hamilton et al., 2016). However, recent studies indicate a significant decrease in estimates over the past decades (Holston et al., 2017). Following the financial crisis, newer estimates

place the steady-state real interest rate in the US at around 1.25% p.a. (Kiley, 2020). Considering that these estimates were obtained in the context of an all-time low in rates, which goes against the long-term trend, I use the historical 2% for $\bar{r}$ in the model. Additionally,$\bar{r}$ governs the subjective discount factor β, which is calculated as $\beta = \frac{1}{(1+\bar{r})}^{\frac{1}{12}} = 0.98$, where one time period represents one month.

Elasticity Intertemporal of Substitution (σ) - EIS

The elasticity of intertemporal substitution exhibits substantial uncertainty in the literature. While many assume the highly pathological case of $\sigma = 1$, corresponding to log utility, recent studies suggest a range for the EIS of 0.3 - 0.4, with the upper bound being 0.8 (Havránek, 2015). These estimates are supported by Crump et al. (2022) and Havránek et al. (2015) in a later study of the US economy alone, finding $\sigma \approx 0.5$ and $\sigma \approx 0.594$, respectively. Consequently, I choose the commonly used estimate $\sigma = 0.5$ in the baseline calibration.

Price Elasticity of Individual Demand for Differentiated Goods (ϵ)

The price elasticity of individual demand curves for differentiated goods determines the steady-state markup of price over marginal cost for the hypothetical fully flexible firm. The elasticity of individual demand is negatively related to the firm's net markup $\frac{1}{(\epsilon-1)}$. Given the monopoly pricing distortion, we must have $\epsilon > 1$. I use estimated firm markup to recover the price elasticity of demand. Hall (2018) finds that gross markups are densely distributed between 1 and 1.5 with a mean of 1.3, which is similar to the standard conservative estimate of 1.2 (Traina, 2018), which I use as a baseline. However, markups have risen within the past years to up to 1.7 (De Loecker & Eeckhout, 2017). This wide range of estimates could be due to averaging over inherently heterogeneous consumer-firm interactions across products, industries, geographic areas, or firm sizes, all of which are important drivers of markups.

Frisch Elasticity of Labour Supply (ψ)

The Frisch Elasticity of labor supply is subject to uncertainty in the economic debate. Microeconometric estimates yield a range of [0,0.5], while macroeconomic estimates for DSGE model calibrations tend to be higher $\psi \in [2, 4]$ (Peterman, 2016). Peterman argues that microeconomic estimates rely on estimation exclusively for married males and limit observations to the intensive margin (fluctuations in hours worked per individual), while macroeconomic estimates account for the whole population looking at intensive and extensive margin (fluctuations in individuals employed). Hence, I opt for the macro-based approach also applied by Chetty et al. (2011) and use 2.8 as the baseline Frisch Elasticity.

Elasticity of Marginal Cost to Firm-level Production (η)

The elasticity of marginal cost with respect to firm-level output is given by $\eta = \frac{\alpha}{(1-\alpha)}$. Hence I use the estimated capital share in production α to construct η. There exists a widespread consensus that $\alpha = \frac{1}{3}$ in advanced economies like the US, which is also confirmed in a recent sectoral study by Herrendorf and Valentinyi (2008). Thus I use $\eta = 0.49$ as a baseline.

Calvo Parameter (θ)

The Calvo parameter θ indicates the probability that a firm cannot change prices in a given time period. I use empirical studies estimating the average duration of a price, which is given by $\frac{1}{(1-\theta)}$ in the staggered-price-change Calvo framework[8], to back out the Calvo parameter. Estimates vary widely due to different treatments of temporary sales and substitution effects between goods. Nakamura and Steinsson (2008) argue temporary sales are short-lived and will be reversed in the future. Hence, I chose a duration measure that excludes sales. Because I apply a time-dependent Calvo model with exogenously determined price changes, I

[8]Consider a two-period economy where a price P_0 is set in t=0. In t=1 P_0 changes with probability $(1-\theta)$ and has a duration of 1, or it does not change with probability θ. Probabilities are compounded and in t=2 P_0 changes with probability $(1-\theta)\theta$ and has a duration of 2, or does not change at all throughout with probability θ^2 and has a duration of 3. Then the average duration for P_0 is $(1-\theta)\times 1+\theta(1-\theta)\times 2+\theta^2\times 3 = 1+\theta+\theta^2$. Similarly, for a three-period economy, we will find the average duration to be $\theta+\theta^2+\theta^3$. Extending the argument to $t \to \infty$ the average duration is $\sum_{i=0}^{\infty}\theta^i = \frac{1}{(1-\theta)}$

opt for the inclusion of price changes due to substitution. The respective US median implied duration is 7.4 to 8.7 months (Nakamura & Steinsson, 2008), which yields a range for δ of [0.86, 0.89]. Naturally, including sales should reduce the duration while excluding substitution will increase the duration for a given price change. Thus, I use the upper limit of 8.7 months ($\theta = 0.8851$) as a baseline estimate.

Theoretical Coupon on Government Bonds (δ)

As argued above, the implied duration of government bonds in SS in our model is given by $\frac{1}{(1-\beta\delta)}$, where $(1-\delta)$ can also be interpreted as the fraction of debt being refinanced every period. The US Treasury reports the percentage of debt maturing in under one year as 29.1% (U.S. Department of the Treasury Office of Debt Management, 2023). Since time is measured in months, $(1-\delta) = \frac{0.291}{12}$, i.e., $\delta = 0.9758$ and an average duration of $\frac{1}{(1-\beta\delta)} = 38.67$ which corresponds roughly to a duration of three years and three months. This finding is mostly congruent with the frequently used measure for the average Macaulay duration of domestic debt in the US economy of four years (Organisation for Economic Co-operation and Development, n.d.), which implies $\delta = 0.98$. I use the US treasury estimate as a baseline, also looking at the special cases of $\delta = 0$ and $\delta = 1$.

Habit-Persistence (ζ)

There exists a large disagreement about the empirical extent of habit formation depending on the estimation technique and the type of habit formation. Bouakez et al. (2005) estimate 0.982 for internal multiplicative habits within a maximum likelihood estimation, Ravn et al. (2006) estimate 0.86 for external deep habits. Smets and Wouters (2003) most closely follows the present model specification estimating a mean habit persistence of 0.59 for superficial external habits, which I use as a baseline.

4 Discussion

Simulating a positive 1% shock to the steady-state level of debt (shortfall of tax revenues by 1%), I look at the impulse response functions (IRF) over 48 periods (4 years) when discussing no habit persistence and 200 periods for IRFs with habits. Habit persistence is frequently motivated by matching a hump-shaped consumption response to shocks which is only observable in the long run in the calibrated model, an obvious shortcoming as the forecasting period seems to be too long to draw reliable conclusions. IRFs illustrate the dynamic response of an endogenous variable to the designated shock. Since I use the log-linearized model, the IRFs are interpreted as % deviations from the steady-state level.

A 1% shock to steady-state debt implies a positive fiscal shock, by our definition, a shock that lowers current tax income T_t without increasing T_{t+h} with $h > 0$, such that the decrease cannot be compensated by an adjustment of future taxes. As outlined by the iterated budget constraint, the central bank faces the trade-off between (A) decreasing future ex-ante real interest rates r_{t+h}, or (B) creating a negative ex-post real return today R_t, to force the debt level back to steady-state.

- Channel (A) decreasing r_{t+h} is achieved via expansionary monetary policy: the central bank decreases i_t for multiple periods. By the Fisher equation, this is associated with an equivalent decrease in r_t. However, decreasing i_t has a stimulating effect on the economy in the presence of nominal rigidities inducing a costly positive output gap and persistent inflation.

- Channel (B) reducing R_t is made possible via increasing j_t, which requires increasing i_t and increasing $\mathbb{E}(i_{t+h})$, recalling that today's bond yield j_t is the weighted average of current and future nominal interest rates.

Both mechanisms are closely intertwined, so I will look at some extreme cases to carve out their differences.

4.1 Simple Economy without Habits

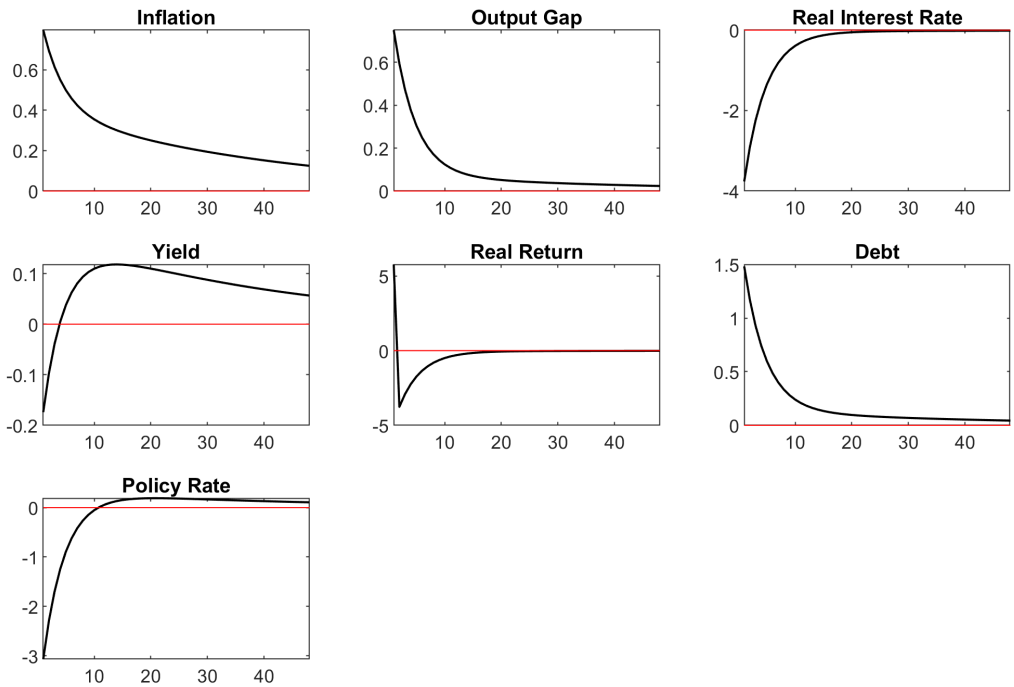

Figure 1: Impulse Response to a 0.01 standard deviation shock of steady-state debt level, for log-linearized model implying a 1% deviation from steady-state debt level. IRFs are shown for 48 periods, one period being one month. Cumulative loss over 200 periods post-shock measured in % deviation from steady-state consumption is 0.29% (manual computation of loss function)

In the baseline case (figure 1): the optimal Ramsey policy introduces an immediate reduction in nominal interest relative to steady-state, associated with a negative yield and real interest rates and positive inflation and output gap relative to steady-state and a short spike up in ex-post real return in the first period, followed by a steep drop in the second period. This indicates that upon impact, the central bank chooses to set off the ex-ante real interest rate channel (A) via expansionary monetary policy by sharply decreasing the policy rate, which induces an immediate drop in the yield due to the expectations theory of the yield. With i_{t_0} falling and $\mathbb{E}(i_{t_0+h})$ remaining below steady-state for the first ten periods, j_t must fall already today. [9] With bond yield and price having an inverse relationship, this

[9]Notice j_t drops by less than the policy rate, which is because the policy rate has a swift return to

implies an immediate upward spike in ex-post real return. The inverse yield-price relationship arises naturally from the fact that by no arbitrage, the nominal bond price must equal the present discounted value of future cash flows, where j_t is the discount rate. With fixed coupon payments, a decreasing yield requires an increase in prices for this to hold. Likewise, the ex-post real rate of return on bonds increased, which is the actual realized return from holding the bond accounting for inflation. From a financial perspective: bondholders locked in a higher YTM when purchasing the bond vis-à-vis the current yield on newly issued bonds. If bondholders sold their bond today before full redemption, they could sell at a higher price, realizing capital gains. The actual ex-post real return is thus determined by said yield-price factors and inflation. Since inflation increases, ex-post real return should decrease in the absence of yield changes. However, with a decreasing yield, ex-post real return spikes because the impact of the yield change is manyfold the impact of inflation (EQ 23). This is actually counterproductive for depressing the real value of debt, even exacerbating the fiscal shock, which spikes 0.48 percentage points more than the shock. [10] Meanwhile, the expansionary monetary policy is instrumental in tackling the shock because it discourages saving and encourages consumption, which motivates a positive inflation and output gap. By the IS equation, the output gap translates into a negative deviation from steady-state for the ex-ante real interest rate. Having established why ex-post real return increases at first, why does it decrease from the second period onward? Again given the inverse yield-price relationship. A quick lift-off for expansionary monetary policy allows a step increase in the policy rate from the second period onward, mapping into increasing yield. Then the yield differential $(j_t - j_{t-1})$ in the yield equation (23) is positive, requiring a decreasing real return. Over time, the less expansionary monetary policy induces a return of the real interest rate and ex-post real return to steady-state. This allows inflation and the output gap to gradually return to steady-state though inflation persistently remains slightly

steady-state in the following period. Hence, the YTM being a weighted average thereof, anticipates the steep rise in i_t in the future

[10] This is precisely because of the increasing YTM, which makes outstanding debt more attractive compared to other investment opportunities, thus increasing its real value

higher even after 100 periods. Hence, persistently low inflation does not seem to be very costly yet instrumental in suppressing the real value of debt.

4.2 Simple Economy with Habit Persistence

The introduction of external consumption habits (figure 2) generates the desired-hump-shaped consumption and output response. As habits are more persistent, the "hump" in output is more stretched out, implying increasingly persistent deviations from steady-state.

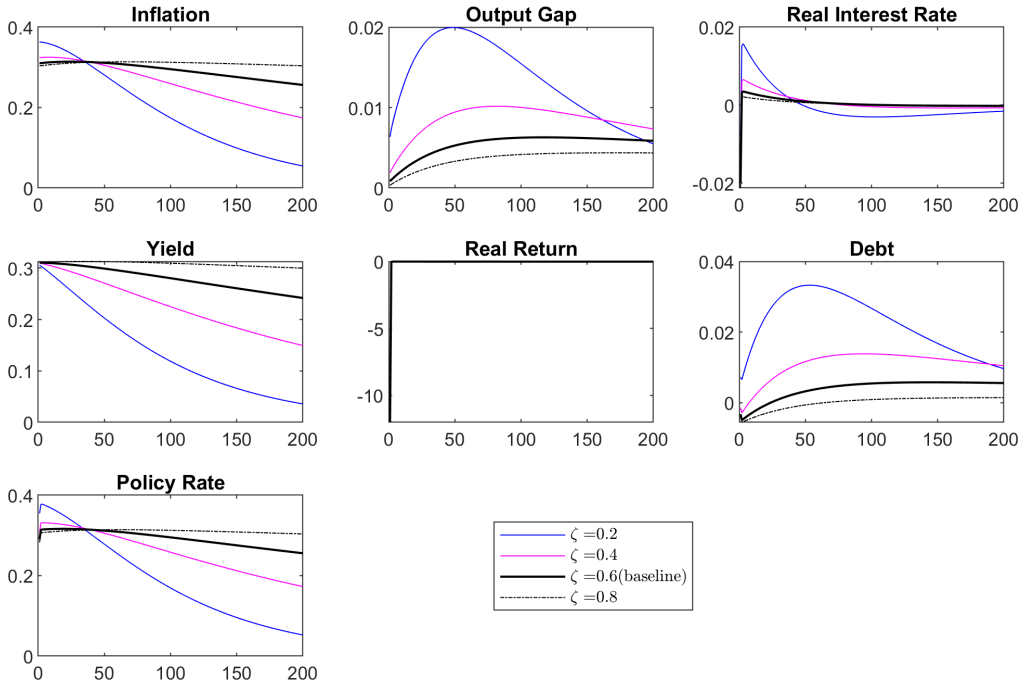

Figure 2: Impulse Response to a 0.01 standard deviation shock of steady-state debt level, for log-linearized model implying a 1% deviation from steady-state debt level. IRFs are shown for 200 periods. Cumulative loss over 200 periods post-shock measured in % deviation from steady-state consumption is 0.67% ($\zeta = 0.2$); 1.21% ($\zeta = 0.4$), 1.94% ($\zeta = 0.6$), 4.55% ($\zeta = 0.8$).

The ex-ante real interest rate channel (A) is used less, and ex-post real return (B) is used more as ζ increases, vis-à-vis the baseline without habits. On the one hand, under external habits, output deviations in the loss function are more costly relative to inflation because current deviations persist, resulting in higher future deviations, thus disincentivizing channel

(A). On the other hand, the lever on ex-ante real interest rates via changing output, governed by the IS curve, is weakened, rendering channel (A) less potent. Previously, central bankers could exploit the interest elasticity of output. In the adjusted IS, this link is watered away by habits, effectively disabling the ex-ante real interest rate channel. Indeed for the baseline $\zeta = 0.6$ the output gap barely moves, deviating from steady-state by a maximum of 0.0033% and real interest rates moving by -0.0201% on impact. Exploiting the ex-post real return channel (B) requires a contractionary monetary policy. Given the Fisher equation, when real interest rates are approximately fixed while nominal interest changes, the price level must adjust to equilibrate the no-arbitrage condition. With positive deviations for yield and inflation, ex-post real return sharply drops, immediately compensating for the debt shock. In the baseline, the ex-post real interest rate drops by -12.02% upon impact in response to a mild 0.29% nominal interest increase. In the subsequent period, real return recovers almost fully to its steady-state level, given the anticipation of no further shocks. As opposed to the no-habit baseline, even small deviations in i_t are sufficient to introduce the desired effect on R_{t0}. A small positive yield has large immediate effects on ex-post real return because current bond-holders face higher returns on newly issued bonds in the long run and are willing to sell bond holdings at a much lower ex-post return to free up capital for reinvestment. Despite full short-run shock compensation, a persistently higher policy rate is necessary to induce the desired movement in the yield given the expectations theory of the yield. A more erratic but short policy reaction is not desirable under the optimal Ramsey policy as persistence in the Phillips and IS curve decelerates the return to steady-state, so the central bank prefers small persistent steady-state deviations over large short-run reactions. As habits become more persistent, this mechanism is more pronounced, and the central bank opts for minuscule changes in the return path for policy rate, yield, and inflation to steady-state. Curiously, the debt level also follows a hump-shaped path. Ex-post real return immediately rebounds to near zero deviation one period after the shock. However, with persistent positive but decreasing inflation and yield, R_t implicitly also follows a hump-shaped path with 0.0032%

deviation in $t = 2$ declining to steady-state from above. This mirrors a hump-shaped debt value simply because of the way how the policy reaction propagates.

4.3 Relevance of the Macaulay Duration

Figure 3 illustrates welfare loss as a function of the Macaulay duration, showing that loss is minimized in the extremes where $\delta = 0$ and $\delta = 1$. With no habit formation, cumulative discounted loss over 200 periods for a 1% fiscal shock is 0.0039% when $\delta = 0$ and 0.0037% when $\delta = 1$. With habit formation at $\zeta = 0.6$, near zero loss is still attainable at the top with 0.0021% loss at $\delta = 1$, but the loss is higher at $\delta = 0$ with 8.89% over 200 periods.

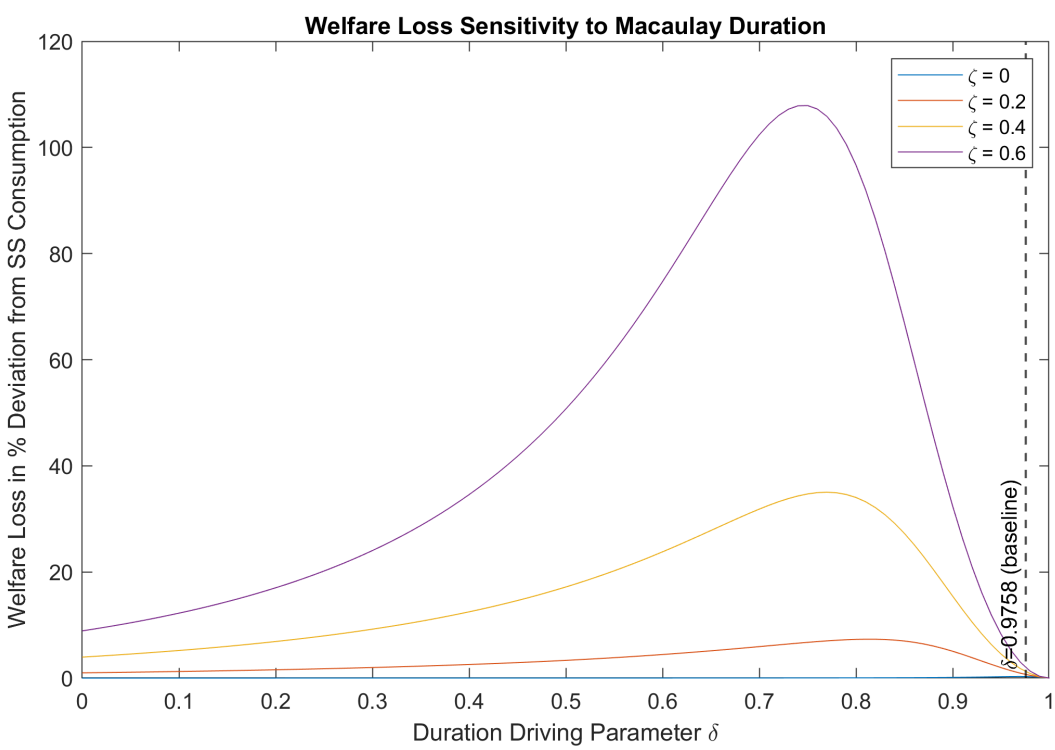

Figure 3: Sensitivity of welfare loss to Macaulay duration of public debt for different levels of habit persistence. Welfare loss is calculated as % deviation from steady-state consumption over 200 periods in response to a 0.01 standard deviation shock of steady-state debt level, for log-linearized model implying a 1% deviation from steady-state debt level.

Periodic Rollover

The case of $\delta = 0$ represents the standard NK model with a periodic rollover of all bonds. Without external habits, the model becomes highly unrealistic: In the absence of persistence, the economy swiftly returns to steady-state even after highly destructive extreme policy choices reflecting almost no loss. The ex-ante real interest rate channel is now the least-costly form of financial repression, while the ex-post real interest-rate channel becomes infeasibly costly and is never used. To see this, setting $\delta = 0$ reduces equation 22 with 23 to $R_t = i_t - \pi_t$. Clearly, a decrease in R_t maps one-for-one into an increase in inflation, which is not desirable given that loss from π weighs orders of magnitude higher than y. Instead, when using the ex-ante real interest channel, yield and nominal interest are driven into the negative deviation. The government can roll over all debt immediately, and the shock gets fully neutralized. Under habit persistence, this loss value is not attainable anymore due to the backward-looking IS and PC curve. The optimal policy response when $\zeta = 0.6$ is even more extreme, with the central bank setting an even lower nominal rate vis-à-vis the baseline. Thus the previous full roll-over approach is equally exploited, with the only difference being an overcompensation of the shock and subsequent return to steady-state with a lag. However, unlike before, inflation remains persistently above steady-state even after 200 periods. Curiously, it is not inflation driving the loss under habit formation but the positive output gap. Under the optimal Ramsey policy, channel (A) is still optimal because inflation is costly, but the output gap weighs more heavily in the loss function, such that loss from deviating from the optimal consumption path is unavoidable, making an almost zero loss unattainable. Another shortcoming of the model with short-term debt is consequently the proposition of radical declines in both the habit-persistent and non-persistent scenario which becomes problematic in the presence of a nominal zero-lower bound given realistic starting values for i_t.

The Case of Perpetuities

At $\delta = 1$, all bonds are perpetuities that are never rolled over. Now welfare loss is particularly

low with and without external habits. Either way, the preferred Ramsey policy immediately reduces the ex-post real interest rate in response to a shock, thus instantly absorbing it in total and even overcompensating slightly. This is achieved with a persistent positive deviation of nominal interest, yield, inflation, and output from steady-state. Intuitively, the ex-ante real interest rate channel cannot be used since it relies on the government being able to refinance itself at a lower cost when real interest is below steady-state. However, refinancing is not possible with perpetuities. The differential impact of external consumption habits on loss and the IRFs is negligible because, under baseline habits, the optimal Ramsey policy keeps real interest close to zero anyways, so $\delta = 1$ simply imposes the preferred policy lever of the habit-persistence-scenario on the non-habit situation. The resulting Ramsey policy is loss minimizing and even preferred over period-rollover despite persistent inflation and a mildly positive output gap because, first, both deviate from steady-state by a minuscule amount. Second, the output gap is increasing only very slowly, the impact of which is compensated by discounting future losses.

5 Conclusion

This essay calibrates a new Keynesian model and quantitatively estimates the welfare implications of the optimal Ramsey Policy for financial repression in the face of a one-off fiscal shock. The model features nominal and real rigidities caused by monopolistic competition and sticky prices in addition to persistence, which is introduced via long-term government bonds and external consumption habits. For plausible calibration parameters, this essay finds that the loss associated even with a small shock of 1% of steady state debt level can be large, indicating that financial repression, in general, may be very costly. For the baseline calibration in the absence of habits, a minimum loss of 0.29% of steady-state consumption can be achieved where the central bank chooses to induce a contractionary monetary policy and persistently low inflation as well as a positive output gap to bring real interest below

steady-steady state allowing the government to roll-over debt at a lower cost to neutralize the shock. However, this channel has limited efficacy and is ruled out in the more realistic scenario of external habits. Even for small habit persistence, the link between real interest and output is weakened, requiring a prohibitively distortionary output gap, which is costly as desired individual consumption is linked to past consumption. Instead, the central bank ex-post reduces bondholder returns. The results suggest first that financial repression is costly and thus should be weighed carefully against alternative policies. Additionally, the optimal Ramsey policy is very sensitive to the choice of parameters, particularly the level of habit persistence. Hence, the model should be interpreted with caution, and future research could aim at constructing a more elaborate version to assess what is driving the lack of robustness of results.

Appendix A: Derivation Linearized Model

Equilibrium Conditions

We have the following key equations where X_t is the habit-adjusted consumption aggregate over households k: Government:

$$T_t + \frac{Q_t B_t}{P_t} = \frac{(1+\delta Q_t)B_t}{P_t} \tag{A1}$$

Households:

$$X_t^{-1/\sigma} = \beta E_t \left(\frac{(1+\delta Q_{t+1})P_t}{P_{t+1}Q_t} X_{t+1}^{-1/\sigma} \right) \tag{A2}$$

$$X_t^{-1/\sigma} = \beta E_t \left(\frac{P_t}{P_{t+1}Q_t} X_{t+1}^{-1/\sigma} \right) \tag{A3}$$

$$Q_t = \frac{1}{1+j_t-\delta} \tag{A4}$$

$$Q_t = \frac{1}{1+i_t} \tag{A5}$$

$$(1+E_t R_{t+1}) = \frac{(1+\delta Q_{t+1})P_t}{Q_t P_{t+1}} = \frac{(1+j_t)}{(1+\pi_{t+1})} \frac{(1-\frac{\delta}{(1+j_t)})}{(1-\frac{\delta}{(1+j_{t+1})})} \tag{A6}$$

$$(1+r_t) = \frac{1+i_t}{1+\pi_{t+1}} \tag{A7}$$

$$(1+E_t R_{t+1}) = (1+r_t) \tag{A8}$$

Firms:

$$P_t = \left((1-\theta) \sum_{l=0}^{\infty} \theta^l \left(P_{t-l}^* \right)^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}} \tag{A9}$$

$$0 = E_t \sum_{l=0}^{\infty} \theta^l q_{t+l|t} Y_{t+l} \left(\frac{P_t^*}{P_{t+l}} \right)^{-\epsilon} \left(\frac{P_t^*}{P_{t+l}} - \mu \left(\frac{P_t^*}{P_{t+l}} \right)^{-\epsilon\eta} z_{t+l} \right) \tag{A10}$$

$$\left(\frac{\widetilde{P}_t}{P_t} \right) = \mu (1+\eta) \frac{w_t}{A_t} \left(\frac{Y_t}{A_t} \right)^{\eta} \left(\frac{\widetilde{P}_t}{P_t} \right)^{-\epsilon\eta} \tag{A11}$$

Equation (A2) is the habit-adjusted consumption euler for $\delta > 0$, (A3) is the equivalent for $\delta = 0$. (A4) and (A5) capture the respective bond prices for $\delta > 0$ and $\delta > 0$. (A6) defines the ex-post real interest rate and (A7) is the standard Fisher Equation.Then equations (A2) and (A6) yield

$$X_t^{-1/\sigma} = \beta E_t((1 + E_t R_{t+1}) X_{t+1}^{-1/\sigma}) \tag{A12}$$

While (A3), (A5), and (A7) together imply

$$X_t^{-1/\sigma} = \beta E_t((1 + r_t) X_{t+1}^{-1/\sigma}) \tag{A13}$$

Together, these imply (A8) as a no-arbitrage condition between short-term and long-term bonds.

I perform a log-linearization around the steady-state where '$^-$' indicates steady-state values of the variable and '$\hat{}$' denotes the deviation from steady-state. Hence $\frac{a_t - \bar{a}}{\bar{a}} \approx \hat{a}_t$ where $\hat{a}_t = log(a_t) - log(\bar{a})$ is the log-deviation from steady-state.

Government Budget Identity

In (A1) w,e can denote the real value of government liabilities at the end of period t as $D_t = \frac{Q_t B_t}{P_t}$, which allows rearranging the government budget identity using the observation that $(1 + \delta Q_t) = \frac{(1+R_t) Q_{t-1} P_t}{P_{t-1}}$

$$T_t + D_t = (1 + R_t) D_{t-1} \tag{A14}$$

We take logs, finding:

$$\ln(D_t + T_t) = \ln(1 + R_t) + \ln(D_{t-1}) \tag{A15}$$

The first-order Taylor approximation around the steady-state and canceling terms yields

$$\frac{1}{\bar{D}+\bar{T}}(D_t-\bar{D})+\frac{1}{\bar{D}+\bar{T}}(T_t-\bar{T})=\frac{1}{\bar{R}}(R_t-\bar{R})+\frac{1}{\bar{D}}(D_{t-1}-\bar{D}) \tag{A16}$$

The the existence of $T_t=\bar{T}$ here is sufficient for a non-explosive debt level where $\bar{T}=\bar{R}\bar{D}$. Then we can rearrange

$$\frac{1}{\bar{D}(1+\bar{R})}(D_t-\bar{D})+\frac{1}{\bar{D}(1+\bar{R})}(T_t-\bar{T})=\frac{1}{\bar{R}}(R_t-\bar{R})+\frac{1}{\bar{D}}(D_{t-1}-\bar{D}) \tag{A17}$$

Now we can denote the log deviation from steady-state debt as d_t and let $\phi_t=\frac{\bar{T}-T_t}{\bar{D}}$ be the size of the fiscal shock relative to steady-state debt. Then also recalling that $\frac{1}{(1+\bar{R})}=\beta$ we finally find the log-linearized government budget constraint

$$\beta\widehat{d_t}=\widehat{d_{t-1}}+\widehat{R}_t+\beta\phi_t \tag{A18}$$

Notice that here we do not distinguish between R_t and its log-deviation $(1+R_t)$.

Bond Market Equations

Log linearizing the no-arbitrage condition (A7) yields the first of three log-linearized bond market equations:

$$\hat{r}_t=E_t\hat{R}_{t+1} \tag{A19}$$

and the equilibrium condition $\bar{R}=\bar{r}$. Taking (A3) we can define $(B_t=\left(\frac{1+\delta Q_{t+1}}{Q_t}\right)=\frac{1+\delta\left(\frac{1}{1+j_{t+1}-\delta}\right)}{\frac{1}{1+j_t-\delta}}$ Then log-linearizing around the steady-state using the rule $a+b\approx\frac{\bar{a}}{\bar{a}+\bar{b}}\hat{a}+\frac{\bar{b}}{\bar{a}+\bar{b}}\hat{b}$, I find

$$\hat{b}_t=\frac{\delta\left(\frac{1}{1+j_{\bar{t}}-\delta}\right)}{1+\delta\left(\frac{1}{1+j_{\bar{t}}-\delta}\right)}\left(-\frac{1+j_{\bar{t}}}{1+j_{\bar{t}}-\delta}\hat{j}_{t+1}\right)+\frac{1+j_{\bar{t}}}{1+j_{\bar{t}}-\delta}\hat{j}_t=-\frac{\delta}{1+j_{\bar{t}}-\delta}E_t\hat{j}_{t+1}+\frac{1+j_{\bar{t}}}{1+j_{\bar{t}}-\delta}\hat{j}_t$$

From (A2) and (A4) I have

$$X_t^{-1/\sigma} = \beta E_t \left(\frac{P_t}{P_{t+1} Q_t} \right) X_{t+1}^{-1/\sigma} = \beta E_t \left(\frac{1+i_t}{1+\pi_{t+1}} \right) X_{t+1}^{-1/\sigma} \tag{A20}$$

Which I rearrange to find

$$1 = \beta E_t \left(\frac{1+\delta Q_{t+1}}{Q_t} \right) \frac{1}{1+\pi_{t+1}} \left(\frac{X_{t+1}}{X_t} \right)^{-1/\sigma} = \beta E_t \left(\frac{1+i_t}{1+\pi_{t+1}} \right) \left(\frac{X_{t+1}}{X_t} \right)^{-1/\sigma} \tag{A21}$$

Then for the log-linearization, this implies $\hat{b}_t = \hat{i}_t$. Recognizing that $\frac{1+j_{\bar{t}}}{1+j_{\bar{t}}-\delta} = \frac{1}{1-\beta\delta}$, I find

$$\hat{i}_t = -\frac{\beta\delta}{1-\beta\delta} E_t \hat{j}_{t+1} + \frac{1}{1-\beta\delta} \hat{j}_t \tag{A22}$$

Rearranging then yields the second bond-market equation for the bond yield

$$\hat{j}_t = \beta\delta E_t \hat{j}_{t+1} + (1-\beta\delta)\hat{i}_t \tag{A23}$$

Finally, substituting the log-linearized Fisher Equation $\hat{i}_t - \hat{\pi}_{t+1} = \hat{R}_{t+1}$ into the log-linearized bond-yield equation reveals the log-linearized price-yield relationship for bonds governing the ex-post real interest rate

$$\hat{R}_t = \hat{j}_{t-1} - \hat{\pi}_t - \frac{\beta\delta}{1-\beta\delta}(\hat{j}_t - \hat{j}_{t-1}) \tag{A24}$$

Habit-adjusted IS Equation

Recall the household utility maximization problem

$$\max_{\{X_{t+l}^k, N_{t+l}^k, B_{t+l}^k\}_{l=0}^{\infty}} \sum_{l=0}^{\infty} \beta^l E_t \left(\frac{(X_{t+l}^k)^{\left(1-\frac{1}{\sigma}\right)} - 1}{1-\frac{1}{\sigma}} - \frac{(N_{t+l}^k)^{\left(1+\frac{1}{\psi}\right)}}{1+\frac{1}{\psi}} \right)$$

subject to

$$C_t^k + \frac{Q_t B_t^k}{P_t} = \frac{W_t N_t^k}{P_t} + \frac{(1+\delta Q_{t+1}) B_{t-1}^k}{P_t} + \frac{D_t^k}{P_t}$$

Taking first-order conditions with respect to B_t and X_t and aggregating over households k

$$\frac{\lambda_t Q_t}{P_t} = \beta E_t \lambda_{t+1} \frac{1+\delta Q_{t+1}}{P_{t+1}} \tag{A25}$$

$$(X_t)^{-\frac{1}{\sigma}} = \lambda_t \tag{A26}$$

Where λ_t is the Lagrange multiplier of the budget constraint, i.e., the shadow value. Then making use of no arbitrage between bonds of different maturities $(1+R_{t+1}) = (1+r_t)$ the FOCs can be log-linearized to obtain:

$$\hat{\lambda}_t = E_t \hat{\lambda}_{t+1} + (\hat{i} - \hat{\pi}_{t+1}) \tag{A27}$$

$$\hat{\lambda}_t = -\frac{1}{\sigma} \hat{X}_t \tag{A28}$$

where recalling the definition of the cross-sectional average consumption aggregate $X_t = C_t - \zeta C_{t-1}$ implies

$$\hat{X}_t = \frac{1}{1-\zeta} \hat{C}_t - \frac{\zeta}{1-\zeta} \hat{C}_{t-1} \tag{A29}$$

Then substituting yields

$$-\frac{1}{\sigma} \left(\frac{(\hat{C}_t) - \zeta(\hat{C}_{t-1})}{1-\zeta} \right) = -\frac{1}{\sigma} E_t \left(\frac{(\hat{C}_{t+1}) - \zeta(\hat{C}_t)}{1-\zeta} \right) + (\hat{i} - \hat{\pi}_{t+1}) \tag{A30}$$

This can be rearranged to the habit-adjusted consumption equation making use of the log-

linearized Fisher equation $\hat{r}_t = \hat{i}_t - \hat{\pi}_{t+1}$:

$$\hat{C}_t = \frac{\zeta}{1+\zeta}\hat{C}_{t-1} + \frac{1}{1+\zeta}E_t(\hat{C}_{t+1}) - \frac{\sigma(1-\zeta)}{1+\zeta}\hat{r}_t \tag{A31}$$

Meanwhile, we can also keep the log-linearized FOC in terms of habit-adjusted aggregate consumption units X_t. Then, for $Y_t = C_t$ and defining $\tilde{Y}_t$ and $\tilde{r}_t$ as the natural level of output and the natural level of interest, respectively, we can define $\tilde{X}_t = \tilde{Y}_t - \zeta\tilde{Y}_{t-1}$,which allows us to write the Euler equation in terms of natural levels:

$$\tilde{X}_t^{-\frac{1}{\sigma}} = \beta E_t(1+\tilde{r}_t)\tilde{X}_t + 1^{-\frac{1}{\sigma}} \tag{A32}$$

And log-linearized:

$$\widehat{\tilde{X}_t} = E_t\widehat{\tilde{X}}_{t+1} - \widehat{\tilde{r}_t} \tag{A33}$$

Now, defining the output gap as $y_t = \frac{Y_t}{\tilde{Y}}$ implies in log-linearized terms $\widehat{\tilde{y}_t} = \widehat{Y}_t - \widehat{\tilde{Y}}_t$. Then, equivalently letting $\widehat{\tilde{x}_t} = \widehat{X}_t - \widehat{\tilde{X}}_t$ allows us to write:

$$\widehat{\tilde{x}_t} = E_t\widehat{\tilde{x}}_{t+1} - \sigma\left(\widehat{r}_t - \widehat{\tilde{r}_t}\right) \tag{A34}$$

And noticing that:

$$\widehat{\tilde{x}_t} = \widehat{X_t} - \widehat{\tilde{X}}_t = \frac{(\widehat{Y}_t - \zeta\widehat{Y}_{t-1}) - (\widehat{\tilde{Y}}_t - \zeta\widehat{\tilde{Y}}_{t-1})}{1-\zeta} = \frac{\widehat{\tilde{y}_t} - \zeta\widehat{\tilde{y}_{t-1}}}{1-\zeta} \tag{A35}$$

Finally, assuming no fluctuations in the natural interest rate, the IS equation becomes:

$$\widehat{\widetilde{y}}_t = \frac{\zeta}{1+\zeta}\widehat{\widetilde{y}}_{t-1} + \frac{1}{1+\zeta}E_t\widehat{\widetilde{y}}_{t+1} - \frac{\sigma(1-\zeta)}{1+\zeta}\widehat{r}_t \tag{A36}$$

Here, the natural output level is defined as the equilibrium output level under flexible prices. This is pinned down by the labor market clearing optimal wage and the optimization of the fully flexible firm, where there are no price distortions $\Delta_t = 1$, given full flexibility:

$$1 = \mu(1+\eta)\frac{\widehat{\widetilde{Y}}^{\left(\frac{1+\eta}{\psi}+\eta\right)}}{A_t^{1+\frac{1+\eta}{\psi}+\eta)}} \tag{A37}$$

And the natural interest rate is the risk-free real interest rate under natural output, pinned down by the Euler equation:

$$\widetilde{r}_t = \left(\frac{\widetilde{Y}_t - \zeta\widetilde{Y}_{t-1}}{\beta E_t(\widetilde{Y}_{t+1} - \zeta\widetilde{Y}_t)}\right)^{\frac{1}{\sigma}} - 1 \tag{A38}$$

Habit-adjusted Phillips Curve

The derivation of the Phillips Curve is largely unaffected by the introduction of habit formation to the degree of how it governs the link between current and future inflation and marginal costs under Calvo pricing. Hence, I provide an abbreviated derivation of Galí (2008) for the Phillips Curve. The central difference arises from the link between real marginal cost and the output gap, which is changed under external habit formation since households' labor supply choice is now dynamic, resulting in a dynamic optimal wage rate.

The log-linearized fully flexible price is given by:

$$\widehat{\widetilde{P}}_t = \widehat{P}_t + \nu\widehat{z}_t \tag{A39}$$

where $\nu = \frac{1}{1+\epsilon\eta}$ is inversely related to firm-level real rigidities and $\widehat{z}_t$ is the log-linearized marginal cost.

The log-linearized price-level equation is given by:

$$\widehat{P}_t = (1-\theta)\sum_{l=0}^{\infty}\theta^l \widehat{P}^*_{t-l} \tag{A40}$$

written recursively as $\widehat{P}_t = \theta\widehat{P}_{t-1} + (1-\theta)\widehat{P}^*_t$. Similarly, the log-linearized reset price equation is found to be:

$$\widehat{P}^*_t = (1-\beta\theta)\sum_{l=0}^{\infty}(\beta\theta)^l E_t(\widehat{P}^*_{t+l} + \nu\widehat{z}_{t+l}) \tag{A41}$$

written recursively as $\widehat{P}^*_t = \beta\theta E_t(\widehat{P}^*_{t+l}) + (1-\beta\theta)\widetilde{P}^*_t$. Substituting the three together yields:

$$\widehat{\pi}_t = \beta E_t(\widehat{\pi}_{t+1}) + \nu\xi\widehat{z}_t \tag{A42}$$

where $\xi = \frac{(1-\beta\theta)(1-\theta)}{\theta}$ is inversely related to nominal rigidities. Now, under external habit formation, the real marginal cost of the flexible firm is:

$$z_t = \frac{(1+\eta)w_t}{A_t}\left(\frac{Y_t}{A_t}\right)^{\eta} \tag{A43}$$

with $w_t = \left(\frac{Y_t - \zeta Y_{t-1}}{A_t}\right)^{1/\sigma}\left(\frac{Y_t}{A_t}\right)^{\frac{1+\eta}{\psi}}$ Thus, log-linearizing delivers:

$$\widehat{z}_t = \left(\frac{(1+\eta)/\psi + \eta}{Y_t}\right)\widehat{Y}_t + \frac{1}{\sigma}\left(\frac{\widehat{Y}_t - \zeta\widehat{Y}_{t-1}}{1-\zeta}\right) - \left(1 + \frac{(1+\eta)/\psi + \eta}{Y_t}\right)\widehat{A}_t \tag{A44}$$

Recalling the equation for natural output $\widetilde{Y}_t$, we can rearrange and log-linearize to write $\widehat{A}_t$ in terms of $\hat{\widetilde{Y}}_t$ and $\hat{\widetilde{Y}}_{t-1}$:

$$\left(1+\frac{1+\eta}{\psi}+\eta\right)\widehat{A}_t=\left(\frac{1+\eta}{\psi}+\eta\right)\hat{\widetilde{Y}}_t+\frac{1}{\sigma}\frac{\widehat{\widetilde{Y}}_t-\zeta\widehat{\widetilde{Y}}_{t-1}}{1-\zeta} \tag{A45}$$

Then $\widehat{z}_t$ can be written in terms of current and past output gaps $\widehat{y}_t$ and $\widehat{y}_{t-1}$, which, after substitution, yields the habit-adjusted Phillips Curve:

$$\widehat{\pi}_t=\beta E_t\widehat{\pi}_{t+1}+\nu\xi\left[\left(\frac{1+\eta}{\psi}+\eta\right)\widehat{y}_t+\frac{\widehat{y}_t-\zeta\widehat{y}_{t-1}}{\sigma(1-\zeta)}\right] \tag{A46}$$

For ease of reading in the main text, I drop the '$\widehat{}$' notation in the linearized system of equations.

Appendix B: Social Planner Problem

Writing the loss function in terms of the usual gap notation is not trivial and requires solving the social planner problem first since in an efficient steady-state, the natural output is equal to the efficient level of output. The social planner is fully in control of the habit-adjusted consumption aggregate X_t, but in addition, also internalizes the choice of the external component of habits. Hence, utility maximization subject to resource constraints amounts to

$$\max_{\{C_{t+l},X_{t+l},N_{t+l}\}_{l=0}^{\infty}}\sum_{l=0}^{\infty}\beta^l E_t\left(\frac{X_{t+l}^{1-1/\sigma}-1}{1-1/\sigma}-\frac{N_{t+l}^{1+1/\psi}}{1+1/\psi}\right)$$

subject to:

$$Y_t=A_tN_t^{1/(1+\eta)}$$

$$C_t=Y_t$$

$$X_t=C_t-\zeta C_{t-1}$$

The Lagrangian of the optimization problem is

$$\Lambda_t = \frac{X_{t+l}^{1-1/\sigma} - 1}{1 - 1/\sigma} - \frac{N_{t+l}^{1+1/\psi}}{1 + 1/\psi} - \gamma_t(C_t - A_t N_t^{1/(1+\eta)}) - \mu_t(X_t - C_t + \zeta C_{t-1})$$

Taking first-order conditions with respect to N_t, C_t, and X_t yields

$$N_t^{1/\psi} = \frac{\gamma_t}{1+\eta} A_t N_t^{-\eta/(1+\eta)} \tag{B1}$$

$$\gamma_t = \mu_t - \beta\zeta E_t \mu_{t+1} \tag{B2}$$

$$X_t^{-1/\sigma} = \gamma_t \tag{B3}$$

Combining the three yields the optimality condition

$$\frac{N_t^{1/\psi}}{X_t^{-1/\sigma}} = \frac{A_t}{1+\eta} N_t^{-\eta/(1+\eta)} \left(1 - \beta\zeta E_t \left(\frac{X_{t+1}}{X_t}\right)^{-1/\sigma}\right) \tag{B4}$$

Which log-linearizes to

$$\left(\frac{1}{\psi} + \frac{\eta}{1+\eta}\right) \widehat{N}_t = \widehat{A}_t - \frac{1}{\sigma}\left(\widehat{X}_t - \beta\zeta \widehat{X}_{t+1}\right) \tag{B5}$$

Likewise, the log-linearized constraints yield

$$\widehat{Y}_t = \frac{1}{1+\eta} \widehat{N}_t \tag{B6}$$

$$\widehat{C}_t = \widehat{Y}_t \tag{B7}$$

$$\widehat{X}_t = \frac{\widehat{C}_t - \zeta \widehat{C}_{t-1}}{1 - \zeta} \tag{B8}$$

Appendix C: Utility-based Welfare

Welfare loss is measured by a utility-based welfare measure that rests on a second-order approximation of utility around the efficient steady-state. I follow Galí (2008), Woodford (2003), and Leith et al. (2012) for the derivation.

For the approximation, I make use of the commonly assumed approximation rule where $\frac{(a_t-\bar{a})}{\bar{a}} \approx \widehat{a}_t + \frac{1}{2}\widehat{a}_t^2$, such that $\widehat{a}_t = \log(a_t) - \log(\bar{a})$ is the deviation from steady-state. The period-t utility of the cross-sectional average household is

$$U_t = \frac{X_t^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} - \frac{N_t^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}}$$

with the habit-adjusted aggregate consumption $X_t = C_t - \zeta C_{t-1}$. Then the second-order Taylor approximation yields

$$U_t - U \approx U_X X \left(\frac{(X_t - X)}{X}\right) + U_N N \left(\frac{(N_t - N)}{N}\right) + \frac{1}{2} U_{XX} X^2 \left(\frac{(X_t - X)}{X}\right)^2 + \frac{1}{2} U_{NN} N^2 \left(\frac{(N_t - N)}{N}\right)^2$$

Written as log-deviations

$$\frac{X_t^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} \approx \bar{X}^{1-\frac{1}{\sigma}} \left(\widehat{X}_t + \frac{1-\frac{1}{\sigma}}{2}\widehat{X}_t^2\right) \tag{C1}$$

$$\frac{N_t^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}} \approx \bar{N}^{1+\frac{1}{\psi}} \left(\widehat{N}_t + \frac{1+\frac{1}{\psi}}{2}\widehat{N}_t^2\right) \tag{C2}$$

Where I make use of the CES-type additively separable utility property $\frac{1}{\sigma} = -\frac{U_{XX}}{U_X}X$ and $-\frac{1}{\psi} = -\frac{U_{NN}}{U_N}N$. The second-order approximation is then best performed separately for consumption and labour.

With external additive habits, $X_t = C_t - \zeta C_{t-1}$ approximates to $X_t \approx \widehat{X}_t + \frac{1}{2}\widehat{X}_t^2$, such that

rearranging implies

$$\widehat{X}_t = \frac{\widehat{C}_t}{1-\zeta} - \frac{\zeta\widehat{C}_{t-1}}{1-\zeta} = \frac{1}{1-\zeta}\left(\widehat{C}_t + \frac{1}{2}\widehat{C}_t^2\right) - \frac{\zeta}{1-\zeta}\left(\widehat{C}_{t-1} + \frac{1}{2}\widehat{C}_{t-1}^2\right) - \frac{1}{2}\widehat{X}_t^2 \tag{C3}$$

And to the second order

$$\widehat{X}_t^2 = \frac{1}{(1-\zeta)^2}(\widehat{C}_t - \zeta\widehat{C}_{t-1})^2 \tag{C4}$$

Together, this implies

$$\left(\widehat{X}_t + \frac{1-\frac{1}{\sigma}}{2}\widehat{X}_t^2\right) = \frac{1}{1-\zeta}\left(\widehat{C}_t + \frac{1}{2}\widehat{C}_t^2\right) - \frac{\zeta}{1-\zeta}\left(\widehat{C}_{t-1} + \frac{1}{2}\widehat{C}_{t-1}^2\right) - \frac{1}{2\sigma}\widehat{X}_t^2 \tag{C5}$$

Then aggregating yields

$$\sum_{l=0}^{\infty}\beta^l E_t\left(\widehat{X}_t + \frac{1-\frac{1}{\sigma}}{2}\widehat{X}_t^2\right) = \sum_{l=0}^{\infty}\beta^l E_t\left(\frac{(1-\beta\zeta)}{(1-\zeta)}\left(\widehat{C}_t + \frac{1}{2}\widehat{C}_t^2\right) - \frac{1}{2\sigma}\widehat{X}_t^2\right) \tag{C6}$$

For the disutility of labor, recall the labor market clearing solution

$$N_t = \left(\frac{Y_t}{A_t\Delta_t}\right)^{1+\eta} \text{ where } \Delta_t = \left(\int_0^1\left(\frac{P_t(\iota)}{P_t}\right)^{-\epsilon(1+\eta)}d\iota\right)^{\frac{1}{1+\eta}} \tag{C7}$$

This implies

$$\widehat{N}_t = (1+\eta)(\widehat{Y}_t + \widehat{A}_t) + V_t \tag{C8}$$

Where I define

$$V_t = \frac{1}{1+\eta}\log\left(\int_0^1\left(\frac{P_t(\iota)}{P_t}\right)^{-\epsilon(1+\eta)}d\iota\right) \tag{C9}$$

Galí (2008) proves that in the neighborhood of a symmetric steady-state, up to a second-

order approximation V_t is proportional to the variance of prices $p_t(\iota)$

$$V_t \approx \frac{\epsilon}{2v}\text{var}_\iota\{p_t(\iota)\} \tag{C10}$$

With $v = \frac{1-\alpha}{1-\alpha+\alpha\epsilon} = \frac{1}{1+\epsilon\eta}$ recalling the definition $1+\eta = \frac{1}{1+\alpha}$. Then for the period-t labor disutility, we have

$$\left(\widehat{N}_t + \frac{1+\frac{1}{\psi}}{2}\widehat{N}_t^2\right) \approx (1+\eta)\left(\widehat{Y}_t + \frac{\epsilon}{2v}\text{var}_\iota\{p_t(\iota)\} + (1+\eta)\frac{1+\frac{1}{\psi}}{2}(\widehat{Y}_t - \widehat{A}_t)^2\right) + t.i.p. \tag{C11}$$

The term (t.i.p.) denotes "terms independent of policy," that is, terms that remain unaffected by the policy choice and, due to their additive nature, drop out in the optimization. Thus, I can summarize them as (t.i.p.). Again, aggregating yields

$$\sum_{l=0}^{\infty}\beta^l\mathbb{E}_t\left(\widehat{N}_t + \frac{1+\frac{1}{\psi}}{2}\widehat{N}_t^2\right) = \sum_{l=0}^{\infty}\beta^l\mathbb{E}_t\,(1+\eta)\left\{\widehat{Y}_t + \frac{\epsilon}{2v}\text{var}_i\left\{p_t(i)\right\} + \frac{1+\eta}{2}\left(\widehat{Y}_t - \widehat{A}_t\right)^2\right\} + t.i.p. \tag{C12}$$

Hence, per-period utility, when making use of market clearing so $\widehat{C}_t = \widehat{Y}_t$, is given by:

$$\begin{aligned}
L_t &= \sum_{l=0}^{\infty}\beta^l\mathbb{E}_t\left(\frac{X_t^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} - \frac{N_t^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}}\right)\\
&= \overline{X}^{1-\frac{1}{\sigma}}\sum_{l=0}^{\infty}\beta^l\mathbb{E}_t\left(\frac{(1-\beta\zeta)}{(1-\zeta)}\left(\widehat{Y}_t + \frac{1}{2}\widehat{Y}_t^2\right) - \frac{1}{2\sigma}\widehat{X}_t^2\right)\\
&\quad - (1+\eta)\overline{N}^{1+\frac{1}{\psi}}\sum_{l=0}^{\infty}\beta^l\mathbb{E}_t\left(\widehat{Y}_t + \frac{\epsilon}{2v}\text{var}_i\left\{p_t(i)\right\} + \frac{1+\eta}{2}\left(\widehat{Y}_t - \widehat{A}_t\right)^2\right) + t.i.p.\\
&= \overline{N}^{1+\frac{1}{\psi}}\sum_{l=0}^{\infty}\beta^l\mathbb{E}_t\left(\left(\widehat{Y}_t + \frac{1}{2}\widehat{Y}_t^2\right) - \frac{1}{2\sigma}\frac{(1-\zeta)}{(1-\beta\zeta)}\widehat{X}_t^2 - \widehat{Y}_t - \frac{\epsilon}{2v}\text{var}_i\left\{p_t(i)\right\} - \frac{1+\eta}{2}\left(\widehat{Y}_t - \widehat{A}_t\right)^2\right) + t.i.p.\\
&= -\frac{1}{2}\overline{N}^{1+\frac{1}{\psi}}\sum_{l=0}^{\infty}\beta^l\mathbb{E}_t\left(-\widehat{Y}_t^2 + \frac{1}{\sigma}\frac{(1-\zeta)}{(1-\beta\zeta)}\widehat{X}_t^2 + \frac{\epsilon}{v}\text{var}_i\left\{p_t(i)\right\} + (1+\eta)(1+\frac{1}{\psi})\left(\widehat{Y}_t - \widehat{A}_t\right)^2\right) + t.i.p.
\end{aligned}$$

Notice that the third equality follows from the steady-state conditions where optimally $\bar{X}^{-1/\sigma}(1-\zeta\beta) = \bar{N}^{1/\psi}$ and since $\bar{X} = (1-\zeta)\bar{Y}$, and with A being constant, $\bar{Y} = \frac{1}{1+\eta}\bar{N}$ (from log-linearizing the production function). Then $\bar{X}^{1-1/\sigma}(1-\zeta\beta) = \frac{(1-\zeta)}{(1+\eta)}\bar{N}^{1+1/\psi}$ allows us to rewrite the loss function as above. I additionally drop the level terms.

Woodford (2003) proved that under Calvo pricing, when θ is the Calvo parameter:

$$\sum_{t=0}^{\infty}\beta^t \text{var}_i\{p_t(i)\} = \frac{\theta}{(1-\beta\theta)(1-\theta)}\sum_{t=0}^{\infty}\beta^t\widehat{\pi}_t^2 \tag{C13}$$

Hence, we can write:

$$L_t = -\frac{1}{2}\Omega\sum_{l=0}^{\infty}\beta^l E_t\left\{-\widehat{Y}_t^2 + \frac{1}{\sigma}\frac{(1-\zeta)}{(1-\beta\zeta)}\widehat{X}_t^2 + \frac{\epsilon}{\upsilon\xi}\widehat{\pi}_t^2 + (1+\eta)(1+\frac{1}{\psi})(\widehat{Y}_t - \widehat{A}_t)^2\right\} + t.i.p. \tag{C14}$$

where $\Omega = N^{1+\frac{1}{\psi}}$ and $\xi = \frac{(1-\beta\theta)(1-\theta)}{\theta}$.

Writing the loss function in terms of the usual gap notation is not trivial but can be shown using the results from the social planner problem.

$$\begin{aligned}
L_t = &-\frac{1}{2}\Omega\sum_{l=0}^{\infty}\beta^l\mathbb{E}_t\left\{-\widehat{Y}_t^2 + \frac{1}{\sigma}\frac{(1-\zeta)}{(1-\beta\zeta)}\widehat{X}_t^2 + \frac{\epsilon}{\upsilon\xi}\widehat{\pi}_t^2 + \left(\frac{1+\eta}{\psi}+\eta\right)\widehat{Y}_t^2\right.\\
&\left.-2\widehat{Y}_t\widehat{A}_t\left(1+\frac{1+\eta}{\psi}+\eta\right)\right\} + t.i.p.\\
= &-\frac{1}{2}\Omega\sum_{l=0}^{\infty}\beta^l\mathbb{E}_t\left\{\frac{1}{\sigma}\frac{(1-\zeta)}{(1-\beta\zeta)}\widehat{X}_t^2 + \frac{\epsilon}{\upsilon\xi}\widehat{\pi}_t^2 + \left(\frac{1+\eta}{\psi}+\eta\right)\widehat{Y}_t^2\right.\\
&\left.-2\left(\frac{1+\eta}{\psi}+\eta\right)\widehat{Y}_t\widehat{\widehat{Y}}_t + 2\left(\frac{1+\eta}{\psi}+\eta\right)\widehat{Y}_t\widehat{\widehat{Y}}_t - 2\widehat{Y}_t\widehat{A}_t\left(1+\frac{1+\eta}{\psi}+\eta\right)\right\} + t.i.p.\\
= &-\frac{1}{2}\Omega\sum_{l=0}^{\infty}\beta^l\mathbb{E}_t\left\{\frac{1}{\sigma}\frac{(1-\zeta)}{(1-\beta\zeta)}\widehat{X}_t^2 + \frac{\epsilon}{\upsilon\xi}\widehat{\pi}_t^2 + \left(\frac{1+\eta}{\psi}+\eta\right)(\widehat{Y}_t - \widehat{\widehat{Y}}_t)^2\right.\\
&\left.+2\left(\frac{1+\eta}{\psi}+\eta\right)\widehat{Y}_t\widehat{\widehat{Y}}_t - 2\widehat{Y}_t\widehat{A}_t\left(1+\frac{1+\eta}{\psi}+\eta\right)\right\} + t.i.p.
\end{aligned}$$

Then notice that

$$
\begin{aligned}
&2\left(\frac{1+\eta}{\psi}+\eta\right)\widehat{Y}_t\widehat{\widetilde{Y}}_t - 2\widehat{Y}_t\widehat{A}_t\left(1+\frac{1+\eta}{\psi}+\eta\right)\\
&= -2\widehat{Y}_t\left(\widehat{A}_t\left(1+\frac{1+\eta}{\psi}+\eta\right)-\left(\frac{1+\eta}{\psi}+\eta\right)\widehat{\widetilde{Y}}_t\right)+t.i.p.\\
&= -2\widehat{Y}_t\left(\widehat{A}_t-\left(\frac{1+\eta}{\psi}+\eta\right)(\widehat{\widetilde{Y}}_t-\widehat{A}_t)\right)+t.i.p.\\
&= -2\widehat{Y}_t\left(\widehat{A}_t-\left(\frac{1}{\psi}+\frac{\eta}{1+\eta}\right)\widehat{N}_t\right)+t.i.p. \quad \text{(recalling from the social planner that } \widehat{\widetilde{Y}}_t-\widehat{A}_t=\frac{1}{1+\eta}\widehat{N}_t)\\
&= -\frac{2}{\sigma(1-\beta\zeta)}\widehat{Y}_t(\widehat{\widetilde{X}}_t-\beta\zeta\widehat{\widetilde{X}}_{t+1})+t.i.p. \quad \text{(from the social planner FOCs)}
\end{aligned}
$$

Also, we can write

$$
\widehat{X}_t^2 = (\widehat{X}_t-\widehat{\widetilde{X}}_t)^2+\widehat{X}_t\widehat{\widetilde{X}}_t-\widehat{\widetilde{X}}_t^2 \tag{C15}
$$

Then we find

$$
\begin{aligned}
\sum_{t=0}^{\infty}\beta^t\frac{1}{\sigma}\frac{1-\zeta}{1-\beta\zeta}\widehat{X}_t\widehat{\widetilde{X}}_t &= \frac{1}{\sigma}\frac{1-\zeta}{1-\beta\zeta}\left(\frac{\widehat{Y}_t-\zeta\widehat{Y}_{t-1}}{1-\zeta}\right)\widehat{\widetilde{X}}_t+\frac{\beta}{\sigma}\frac{1-\zeta}{1-\beta\zeta}\left(\frac{\widehat{Y}_{t+1}-\zeta\widehat{Y}_t}{1-\zeta}\right)\widehat{\widetilde{X}}_{t+1}+\cdots\\
&= \frac{1}{\sigma}\frac{1}{1-\beta\zeta}\left(\widehat{Y}_t\widehat{\widetilde{X}}_t-\zeta\widehat{Y}_{t-1}\widehat{\widetilde{X}}_t+\beta\widehat{Y}_{t+1}\widehat{\widetilde{X}}_{t+1}+\beta\zeta\widehat{Y}_t\widehat{\widetilde{X}}_{t+1}+\cdots\right)\\
&= \sum_{t=0}^{\infty}\beta^t\frac{1}{\sigma}\frac{1}{1-\beta\zeta}\widehat{Y}_t\left(\widehat{\widetilde{X}}_t-\beta\zeta\widehat{\widetilde{X}}_{t+1}\right)
\end{aligned}
$$

Hence, together, this implies that the loss function in the usual gap notation becomes

$$
L_t = -\frac{1}{2}\Omega\sum_{l=0}^{\infty}\beta^l E_t\left\{\frac{\epsilon}{v\zeta}\widehat{\pi}_t^2+\left(\frac{1+\eta}{\psi}+\eta\right)\left(\widehat{Y}_t-\widehat{\widetilde{Y}}_t\right)^2+\frac{1}{\sigma}\frac{1-\zeta}{1-\beta\zeta}\left(\widehat{X}_t-\widehat{\widetilde{X}}_t\right)^2\right\}+t.i.p.
$$

Finally, since we had $\widehat{x}_t=\widehat{X}_t-\widehat{\widetilde{X}}_t=\frac{\widehat{y}_t-\zeta\widehat{y}_{t-1}}{1-\zeta}$ This implies

$$
\begin{aligned}
L_t &= -\frac{1}{2}\Omega \sum_{l=0}^{\infty} \beta^l E_t \left\{ \frac{1}{\sigma(1-\beta\zeta)(1-\zeta)} + \frac{1+\eta}{\psi} + \eta \right\} \widehat{y}_t^2 - \frac{2\zeta}{\sigma(1-\beta\zeta)(1-\zeta)} \widehat{y}_t \widehat{y}_{t-1} \\
&\quad + \frac{\zeta^2}{\sigma(1-\beta\zeta)(1-\zeta)} \widehat{y}_{t-1}^2 + \frac{\epsilon}{v\zeta} \widehat{\pi}_t^2 \\
&= -\frac{1}{2}\Omega \sum_{l=0}^{\infty} \beta^l E_t \left\{ \frac{\epsilon}{v\zeta} \widehat{\pi}_t^2 + \left(\omega + \frac{1+\eta}{\psi} + \eta \right) \widehat{y}_t^2 - 2\zeta\omega \widehat{y}_t \widehat{y}_{t-1} + \zeta^2 \omega \widehat{y}_{t-1}^2 \right\} + t.i.p.
\end{aligned}
$$

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Miscellaneous Figures

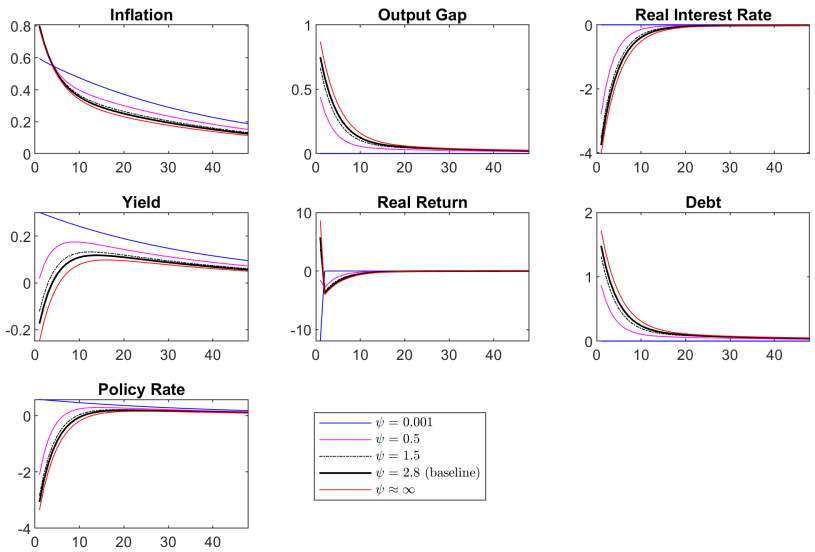

Figure 4: Sensitivity of IRFs in baseline no-habit model to the Frisch elasticity of labour supply following a 0.01 standard deviation shock of steady-state debt level. Cumulative loss over 200 periods post-shock measured in % deviation from steady-state consumption is 0.39% ($\psi = 0.001$), 0.33% ($\psi = 0.5$), 0.30% ($\psi = 1.5$), 0.29% ($\psi = 2.8$ - baseline), 0.27% ($\psi = 1000$),

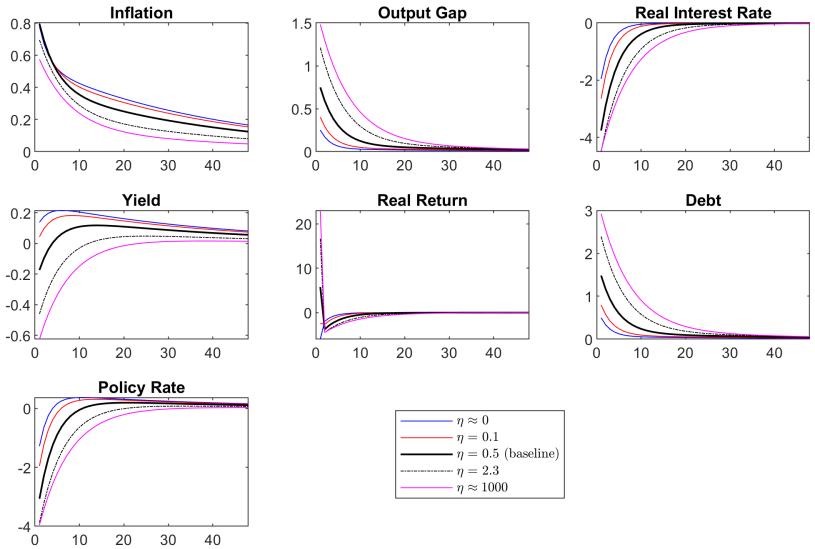

Figure 5: Sensitivity of IRFs in baseline no-habit model to the η the elasticity of marginal cost wrt. firm-level production following a 0.01 standard deviation shock of steady-state debt level. Cumulative loss over 200 periods post-shock measured in % deviation from steady-state consumption is 0.14% ($\eta = 0.1$), 0.29% ($\eta = 0.5$ - baseline), 0.79% ($\eta = 2.3$), 232% ($\eta = 1000$)

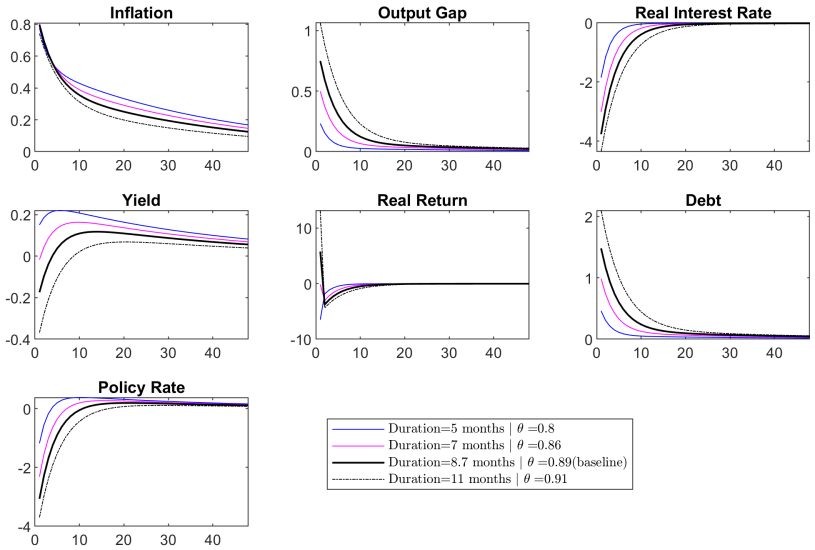

Figure 6: Sensitivity of IRFs in baseline no-habit model to the Calvo price-stickiness parameter following a 0.01 standard deviation shock of steady-state debt level. Cumulative loss over 200 periods post-shock measured in % deviation from steady-state consumption is 0.11% ($\theta = 0.8$), 0.21% ($\theta = 0.86$), 0.29% ($\theta = 0.89$ - baseline), 0.39% ($\theta = 0.91$)

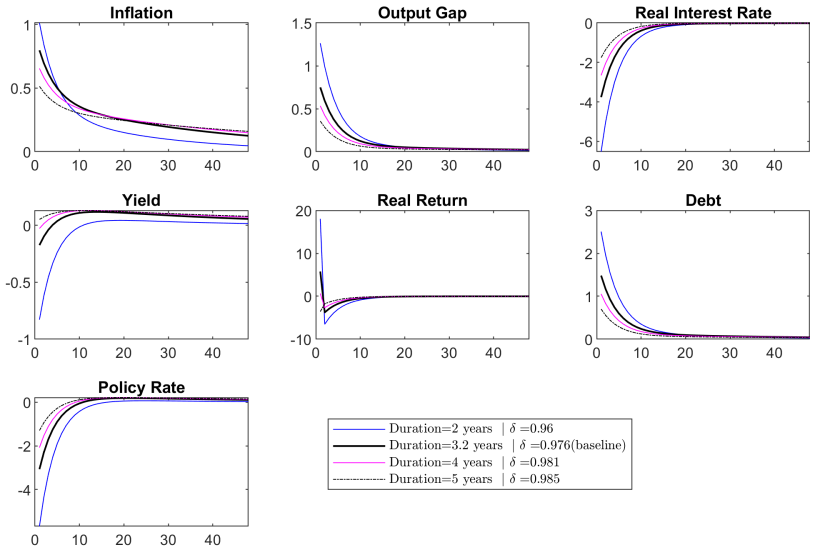

Figure 7: Sensitivity of IRFs in baseline no-habit model to the Macaulay duration of debt following a 0.01 standard deviation shock of steady-state debt level. Cumulative loss over 200 periods post-shock measured in % deviation from steady-state consumption is 0.29% ($\delta = 0.96$), 0.29% ($\delta = 0.976$ - baseline), 0.26% ($\delta = 0.981$), 0.22% ($\delta = 0.985$)